

Cisco Packet Tracer Project #04

PROJECT #04 DHCP & DNS

1. Project Overview

This project simulates a small office network using Cisco Packet Tracer. The network is segmented into two VLANs, IT and FINANCE, to separate departmental traffic.

Router-on-a-Stick is implemented on Cisco Router R1 to enable communication between VLAN 10 and VLAN 20.

The router also provides DHCP services, while SERVER-01 is configured as a DNS server for local hostname resolution.

2. Network Devices

Device	Quantity	Function
Cisco 2911 Router	1	Inter-VLAN Routing & DHCP
Cisco 2960 Switch	1	VLAN & Layer 2 Switching
PC	6	DHCP Client
Server	1	DNS Server

3. VLAN Configuration

VLAN	Name	Network	Gateway
10	IT	192.168.10.0/24	192.168.10.1
20	FINANCE	192.168.20.0/24	192.168.20.1

4. Port Assignment

Port	VLAN	Connected Device
Fa0/1	10	PC-01
Fa0/2	10	PC-02
Fa0/3	10	PC-03
Fa0/4	20	PC-04
Fa0/5	20	PC-05
Fa0/6	20	PC-06
Fa0/7	10	SERVER-01
Fa0/24	Trunk	R1

5. Router-on-a-Stick

VLAN 10

```
interface gigabitEthernet 0/0.10
 encapsulation dot1Q 10
 ip address 192.168.10.1 255.255.255.0
```

VLAN 20

```
interface gigabitEthernet 0/0.20
```

```
encapsulation dot1Q 20
ip address 192.168.20.1 255.255.255.0
```

Physical Interface

```
interface gigabitEthernet 0/0
no shutdown
```

6. DHCP Configuration

IT DHCP Pool

```
ip dhcp excluded-address 192.168.10.1 192.168.10.9
```

```
ip dhcp pool IT_POOL
```

```
network 192.168.10.0 255.255.255.0
```

```
default-router 192.168.10.1
```

```
dns-server 192.168.10.2
```

Finance DHCP Pool

```
ip dhcp excluded-address 192.168.20.1 192.168.20.9
```

```
ip dhcp pool FINANCE_POOL
```

```
network 192.168.20.0 255.255.255.0
```

```
default-router 192.168.20.1
```

```
dns-server 192.168.10.2
```

7. DNS Configuration

SERVER-01

```
IP Address   : 192.168.10.2
```

```
Subnet Mask  : 255.255.255.0
```

```
Gateway      : 192.168.10.1
```

```
DNS Server   : 192.168.10.2
```

DNS Record

```
Hostname     : www.office.local
```

```
IP Address   : 192.168.10.2
```

SERVER-01 is configured with a static IP address and provides DNS services for the local network. The hostname www.office.local resolves to the server's IP address.

8. Network Verification

Switch Verification

```
show vlan brief
```

Trunk Verification

```
show interfaces fa0/24 switchport
```

Router Verification

```
show ip interface brief
```

DHCP Verification
show ip dhcp pool
show ip dhcp binding